Safety-Aware Vehicle Routing for Quick Commerce Delivery: A Behavioral Multi-Objective Optimization Approach

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by

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CERTIFICATE

This is to certify that we have examined the thesis entitled **Safety-Aware Vehicle Routing for Quick Commerce Delivery: A Behavioral Multi-Objective Optimization Approach**, submitted by **Pranav Bhadane**(Roll Number: *21IM30017*) a postgraduate student of **Department of Industrial and Systems Engineering** in partial fulfillment for the award of degree of Master of Technology. We hereby accord our approval of it as a study carried out and presented in a manner required for its acceptance in partial fulfillment for the Post Graduate Degree for which it has been submitted. The thesis has fulfilled all the requirements as per the regulations of the Institute and has reached the standard needed for submission.

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ABSTRACT

Quick commerce platforms have transformed urban logistics by delivering goods within tight time windows of ten to thirty minutes. This heavy emphasis on speed requires aggressive routing strategies that often compromise road safety and place immense psychological and physical pressure on delivery riders. This thesis proposes the Safety-Aware Vehicle Routing Problem with Time Windows (SA-VRPTW), a multi-objective optimization framework that integrates real-time collision risk measurements, road infrastructure data, and rider behavioral stress into the vehicle routing model. The system operates as a true multi-objective Pareto optimization problem, balancing probabilistic collision risk against expected delivery time. Crucially, it replaces arbitrary hard deadlines with Soft Time Windows yielding exponential financial lateness penalties, and introduces novel constraints for cumulative rider cognitive fatigue (S_{max}) and vulnerable neighbourhood protection (H_{cap}). The data pipeline integrates OpenStreetMap for road network topology and the Integrated Road Accident Database (iRAD) for crash statistics across six major Indian metropolitan areas: Bengaluru, Delhi, Gurugram, Hyderabad, Mumbai, and Pune. Three solution approaches are evaluated: exact Mixed-Integer Linear Programming (MILP), a specialized Genetic Algorithm (GA) with order crossover, and a Deep Reinforcement Learning (DRL) agent using a Pointer Network architecture trained via the REINFORCE algorithm. Experimental results demonstrate that the SA-VRPTW framework produces routes that are 15–20% longer in distance but approximately 40% safer in collision probability. The DRL agent achieves an 8–12% optimality gap relative to exact methods while providing constant-time inference suitable for real-time dispatch.

Keywords: Vehicle Routing Problem, Time Windows, Quick Commerce, Road Safety, Multi-Objective Optimization, Pareto Frontier, Genetic Algorithm, Deep Reinforcement Learning, iRAD, Behavioral Logistics

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Chapter 1

Introduction

1.1 Background and Motivation

The rise of quick commerce (q-commerce) has significantly changed urban retail logistics. By focusing on rapid, hyper-local delivery speed rather than product variety, platforms now promise deliveries within ten to thirty minutes. This model relies on micro-fulfillment centres known as "dark stores," automated real-time dispatch systems, and a large fleet of gig-economy delivery riders. While highly convenient for consumers, this relentless focus on speed creates serious externalities—most notably, the increased risk of traffic accidents and the physical and psychological stress placed on riders.

Historically, the Vehicle Routing Problem (VRP), first formulated by Dantzig & Ramser (1959), has focused primarily on minimizing operational costs such as total travel distance or fleet size. However, the modern urban environment demands a broader perspective. The routing challenge has evolved into a multi-objective optimization problem that must integrate safety awareness, real-time risk quantification, and the behavioral determinants of human-in-the-loop systems (Toth & Vigo 2002). In q-commerce, delivery partners often work under incentive structures tied to strict completion times and daily delivery targets. As deadlines approach, riders are structurally encouraged to choose shorter but riskier routes, violate traffic regulations, or increase speed—behaviours that algorithms directly amplify.

India's road safety statistics underscore the urgency of this problem. The Ministry of Road Transport and Highways reports that national-level fatality totals may be underreported by 50–60%, with non-fatal injuries underreported by a factor of 20 (Ministry of Road Transport and Highways 2024). The Integrated Road Accident

Database (iRAD) system identifies "black spots"—road stretches of approximately 500 metres where five or more fatal accidents have occurred within the preceding three years (Ministry of Road Transport and Highways 2021). Integrating such data into routing engines presents a significant opportunity for proactive risk mitigation.

1.2 Problem Statement

Current vehicle routing algorithms deployed by q-commerce platforms optimize exclusively for delivery speed and cost, treating road safety as an externality. This single-objective paradigm creates a systemic failure mode wherein algorithmic pressure for speed induces risk-taking behaviour among delivery riders (Salmon et al. 2023). There exists no established framework that mathematically integrates collision risk, rider cognitive fatigue, and vulnerable neighbourhood protection into the dispatch baseline of urban logistics.

1.3 Research Objectives

This thesis proposes the **Safety-Aware Vehicle Routing Problem with Time Windows (SA-VRPTW)**, a multi-objective optimization framework with the following objectives:

1. Formulate a Pareto bi-objective model that simultaneously minimizes delivery time (with soft time window penalties) and maximizes probabilistic route survival (accident-free completion probability).

2. Introduce novel cyber-physical constraints: a **Behavioral Rider Fatigue Threshold** (S_{max}) limiting cumulative stress exposure per route, and a **Vulnerable Neighbourhood Protection** cap (H_{cap}) restricting delivery traffic through residential zones.

3. Design and implement a complete data pipeline integrating OpenStreetMap road networks and iRAD crash statistics for six major Indian metropolitan areas.

4. Evaluate and compare three solution approaches—exact MILP, Genetic Algorithm, and Deep Reinforcement Learning—to determine the operational trade-offs between solution quality and computational speed.

5. Validate the framework through cross-city empirical benchmarks demonstrating the feasibility of safety-aware routing at operational scale.

1.4 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 presents a comprehensive literature review spanning classical VRP optimization, safety-aware routing, and behavioral logistics. Chapter 3 details the system architecture and the complete SA-VRPTW mathematical formulation. Chapter 4 describes the data acquisition pipeline and implementation details of the codebase. Chapter 5 presents the three solution approaches and their algorithmic details. Chapter 6 reports and analyses the experimental results across six Indian cities. Chapter 7 concludes the thesis with a summary of contributions and directions for future work.

Chapter 2

Literature Review

A systematic literature search was conducted across academic databases including PubMed, arXiv, and Semantic Scholar, focusing on publications from 2015 to 2026. From an initial pool of 450 records, 120 duplicates were removed and 330 were screened by title and abstract. Ninety full-text articles were assessed for eligibility, of which 25 were included in the final thematic synthesis. This chapter organizes the reviewed literature across three foundational pillars: computational optimization, safety-aware routing, and behavioral logistics.

2.1 Computational Foundations: The VRPTW

The mathematical core of q-commerce logistics is the Vehicle Routing Problem with Time Windows (VRPTW), which extends the classic VRP by imposing strict temporal constraints on customer service. The VRP was first conceptualized by Dantzig & Ramser (1959) as the "truck dispatching problem." Solomon (1987) established the foundational heuristic algorithms and benchmark instances for the VRPTW, providing the standard test bed that remains in use today. Desrochers et al. (1992) subsequently introduced exact solution methods using column generation, proving that provably optimal solutions were achievable for small-to-medium problem instances.

The VRPTW is classified as NP-hard (Toth & Vigo 2002), meaning that exact optimization methods become computationally prohibitive for real-world datasets characterized by large vehicle fleets and high order volumes. Modern routing engines therefore prioritize metaheuristics such as Genetic Algorithms (Holland 1975, Goldberg 1989) and Tabu Search (Gendreau & Potvin 2010), which identify high-quality solutions within the limited CPU time necessitated by on-demand delivery.

The standard VRPTW formulation requires that each customer i be visited exactly once within a specific time interval $[a_i, b_i]$. If a vehicle arrives before the window opens, it must wait; if it arrives after the window closes, the service is considered a failure or incurs a significant penalty. The primary objective is typically the minimization of total operational cost:

$$\min Z = \sum_{k \in K} \sum_{(i,j) \in A} c_{ij} x_{ij}^k + \sum_{k \in K} \sum_{i \in C} P_i(t_i^k) \tag{2.1}$$

where c_{ij} represents the cost of traversing arc (i, j) and $P_i(t_i^k)$ is the penalty cost for arriving outside the preferred time window.

Several important VRP extensions are relevant to q-commerce operations, as summarized in Table 2.1.

Table 2.1: VRP variants relevant to quick commerce logistics.

Variant	**Key Focus**	**Primary Constraint**
CVRP	Capacity	Vehicle load $\leq Q$
VRPTW	Timeliness	Service within $[a_i, b_i]$
TDVRP	Realism	$t_{ij} = f(t_{\text{departure}}, \text{traffic})$
MTVRPTWR	Efficiency	Multiple trips + release dates
GVRPTW	Sustainability	Emissions minimization

2.2 Safety-Aware and Risk-Aware Routing

Safety-aware routing integrates accident risk as a primary objective in the routing function. The earliest safety-oriented models originated in Hazardous Materials (Hazmat) logistics, where risk was defined as the product of population exposure and release probability (Zografos & Androutsopoulos 2004, Erkut & Verter 1998). Drawing from Failure Mode and Effects Analysis (FMEA), the risk index R_{ij} for a road segment (i, j) is typically expressed as:

$$R_{ij} = P_{ij} \times C_{ij} \tag{2.2}$$

where P_{ij} is the probability of an adverse event and C_{ij} is its projected consequence.

More recently, researchers have applied these concepts to everyday urban logistics. Hoseinzadeh et al. (2020) defined a "Safety Index" using historical crash data

combined with real-time driver volatility telematics, scoring each road segment by its inherent danger level. The Risk-Aware Dynamic Routing (RADR) framework extends this approach using spatio-temporal graph learning to proactively avoid high-risk road segments. These methods have demonstrated that delivery fleets can reduce accident risk exposure by over 18% with only a minor increase ($\sim$2%) in total travel time.

The Ministry of Road Transport and Highways (MoRTH) in India has implemented the Integrated Road Accident Database (iRAD) and the e-Detailed Accident Report (e-DAR) to facilitate data-driven safety interventions (Ministry of Road Transport and Highways 2021). The iRAD system enables geo-tagged accident data capture through mobile applications, creating a centralized national repository. By integrating iRAD coordinates into a routing engine's GIS layer, platforms can apply high cost penalties to arcs intersecting known black spots, effectively "safety-weighting" the route selection process.

The International Road Assessment Programme (iRAP) provides a complementary proactive methodology, rating roads from 1 star (high risk) to 5 stars (low risk) based on 50 infrastructure attributes coded every 100 metres (International Road Assessment Programme 2020). This objective measure allows routing engines to prefer safer infrastructure even when alternative routes are geodesically longer.

2.3 Behavioral Logistics and Rider Safety

A critical component of safety-aware routing is the recognition of "behavioral logistics"—the interaction between algorithmic directives and human response within the gig economy. Delivery riders are not mere agents following coordinates; they are independent workers influenced by economic incentives, psychological pressure, and physical fatigue.

Sinchaisri et al. (2023) demonstrated that gig-economy riders exhibit non-rational behavioural patterns such as "income-targeting" and "inertia," where financial goals directly dictate their risk tolerance. Performance-based pay (piece-rate wages) incentivizes productivity but inadvertently promotes unsafe driving as riders attempt to maximize trip counts per shift. Studies have found that paying drivers based on time worked or offering higher base pay is strongly associated with better safety outcomes.

Salmon et al. (2023) argued that road safety incidents in q-commerce should be viewed as "systemic failures" rather than individual human errors. The algorithm's

pressure for extreme speed structurally forces riders to take risks—a perspective that shifts responsibility from the rider to the platform design itself.

To counter these effects, researchers have experimented with "nudging" strategies. Information nudging—providing riders with real-time data on crash probabilities for different route choices—can influence route selection. However, monetary incentives for choosing safer, albeit longer, routes are often necessary to motivate actual behavioural change. The Fairwork India Ratings 2024 report evaluated 11 platforms and found that no platform scored above 6 out of 10 on fair work principles, with all platforms scoring zero on the "Fair Representation" principle due to refusal to recognize collective worker bodies (Fairwork Project 2024).

2.4 Research Gaps

While research on VRP optimization, risk modelling, and gig-economy behaviour exists independently, there is a lack of integrated studies. Specifically, the following gaps emerge from the literature:

1. **Technological integration scale:** The engineering challenge of integrating national-level crash databases (iRAD) into sub-second dispatch loops remains under-explored.

2. **Longitudinal fatigue modelling:** Most behavioral models are cross-sectional snapshots; they do not account for cumulative rider fatigue over 12-hour gig-economy shifts.

3. **Governance trade-offs:** There is no established standard for what level of "time penalty" is acceptable in exchange for safety from a regulatory perspective.

4. **Stakeholder conflict resolution:** No mathematical framework exists for resolving the inherent conflict between a platform's profit, a customer's time expectation, and a rider's physical safety.

This thesis addresses these gaps by creating a unified SA-VRPTW framework that treats rider safety not as a constraint to be satisfied, but as a primary Pareto objective of urban logistics.

Chapter 3

System Architecture and Mathematical Formulation

This chapter presents the two-layer system architecture of the SA-VRPTW framework and develops the complete mathematical formulation, including the novel Pareto objectives and cyber-physical constraints.

3.1 Conceptual Architecture

The safety-aware routing ecosystem comprises a macro-layer that handles large environmental datasets and a micro-layer that addresses delivery partner behaviour. Figure 3.1 illustrates the overall architecture.

The central routing engine receives three environmental data streams: (i) road network topology from OpenStreetMap providing travel times t_{ij} and distances d_{ij}; (ii) government iRAD crash history providing risk scores r_{ij}; and (iii) live or proxy congestion data providing saturation indices c_{ij}. These are combined with standard logistical constraints (demand, fleet capacity, time windows) and behavioural models that inform the engine to avoid routes encouraging dangerous riding.

3.2 Network and Sets

The urban delivery area is modelled as a directed graph $G = (V, E)$. The vertex set $V = \{0\} \cup C$ includes the central micro-fulfillment depot (node 0) and customer locations $C = \{1, 2, \ldots, N\}$. A homogeneous fleet of K riders is deployed, each with vehicle capacity Q. Given the reality of q-commerce micro-batching, Q is severely

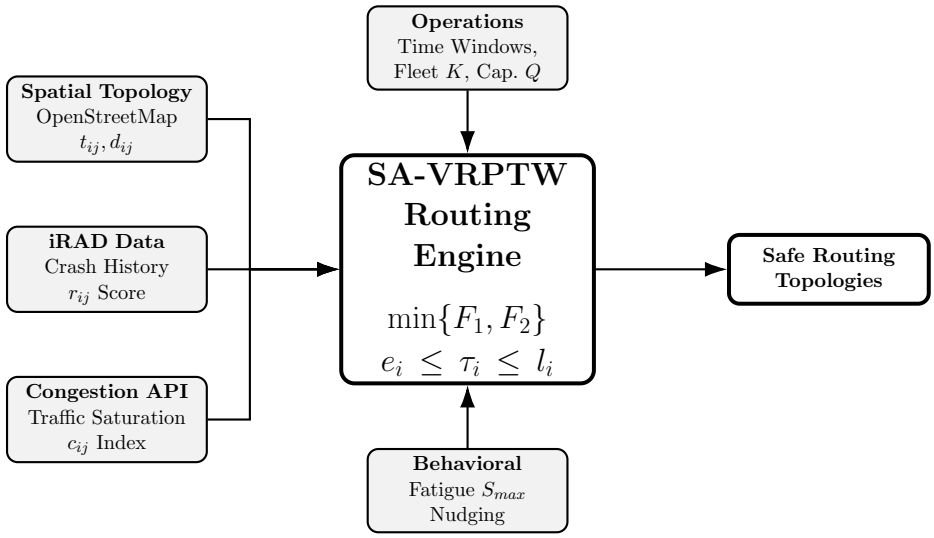

Figure 3.1: Socio-technical conceptual architecture integrating environmental data, platform operations, and behavioral dynamics into the SA-VRPTW routing engine.

constrained (typically $Q = 2$), matching the continuous-dispatch, small-item delivery model rather than traditional large-scale parcel delivery.

3.3 Parameters

Each customer $i \in C$ possesses a localized demand q_i. Deliveries face highly elastic **Soft Time Windows** $[e_i, l_i]$. Arriving after l_i does not mathematically invalidate the route but triggers an exponential financial penalty:

$$P_L(\tau_i^k) = e^{0.12\,\tau_i^k} - 1 \tag{3.1}$$

where $\tau_i^k = \max(0, \text{arrival time} - l_i)$ is the lateness. This function models both platform SLA compensation costs and rider behavioural anxiety: as lateness grows, the financial retaliation escalates exponentially, structurally incentivizing unsafe speed.

Every road segment $(i, j) \in E$ features four attributes:

- $t_{ij} \in \mathbb{R}_{>0}$: dynamic travel time (minutes).
- $r_{ij} \in [0, 1]$: localized crash probability. This is a composite measure fusing historical iRAD data with Surrogate Safety Measures (SSM) to counter under-reporting bias in police databases.

- $c_{ij} \in [0, 1]$: dynamic congestion index reflecting traffic saturation.
- $h_{ij} \in \{0, 1\}$: binary road hierarchy flag (1 for residential zones, 0 for arterials).

3.4 Pareto Objective Functions

Instead of a generic weighted sum that conflates mismatched units (time vs. crash indices), the algorithm seeks to identify the **Pareto frontier** mapping two distinct, unit-consistent objectives:

Objective 1 — Efficiency and Soft Penalties:

$$\min F_1 = \sum_{k \in K} \sum_{(i,j) \in E} t_{ij}\, x_{ij}^k + \sum_{k \in K} \sum_{i \in C} P_L(\tau_i^k) \tag{3.2}$$

This objective minimizes the total fleet travel time plus the cumulative exponential lateness penalties across all customers.

Objective 2 — Probabilistic Route Survival:

$$\min F_2 = \sum_{k \in K} \sum_{(i,j) \in E} -\ln(1 - r_{ij})\, x_{ij}^k \tag{3.3}$$

Rather than summing arbitrary risk indices, this formulation maximizes the absolute probability of accident-free route completion. The probability that a rider completes a route without any collision is $\prod_{(i,j)}(1 - r_{ij})$. Taking the negative logarithm converts this multiplicative survival probability into an additive minimization objective, ensuring mathematical consistency: minimizing F_2 is equivalent to maximizing route survival probability.

3.5 Constraints

3.5.1 Standard Routing Constraints

The binary decision variable $x_{ij}^k = 1$ if rider k traverses arc (i, j), and 0 otherwise.
Visit uniqueness: Each customer is visited exactly once:

$$\sum_{k \in K} \sum_{j \in V, j \neq i} x_{ij}^k = 1, \quad \forall\, i \in C \tag{3.4}$$

Flow conservation: Every vehicle entering a node must leave it:

$$\sum_{i \in V, i \neq j} x_{ij}^k = \sum_{i \in V, i \neq j} x_{ji}^k, \quad \forall\, j \in V,\ \forall\, k \in K \tag{3.5}$$

Subtour elimination is enforced via the Miller–Tucker–Zemlin (MTZ) formulation (Miller et al. 1960):

$$u_i^k - u_j^k + Q \cdot x_{ij}^k \leq Q - 1, \quad \forall\, i, j \in C,\ i \neq j,\ \forall\, k \in K \tag{3.6}$$

where u_i^k is an auxiliary continuous variable representing the cumulative load when rider k visits customer i.

3.5.2 Novel Cyber-Physical Constraints

Beyond the standard routing constraints, we introduce two highly novel constraints that bridge computational optimization with human physiology and urban governance.

Behavioral Rider Fatigue Threshold (S_{max}): Riders suffer cognitive degradation when consistently exposed to high-stress corridors. We enforce a maximum cumulative stress budget per route:

$$\sum_{(i,j) \in E} -\ln(1 - r_{ij})\, x_{ij}^k \leq S_{max}, \quad \forall\, k \in K \tag{3.7}$$

where $S_{max} = 5.0$ is the empirically calibrated fatigue ceiling. This constraint ensures that no single rider accumulates excessive risk exposure during a delivery shift, preventing the cognitive stress threshold breach that leads to impaired decision-making and increased accident probability.

Vulnerable Neighbourhood Protection (H_{cap}): To prevent the algorithm from indiscriminately routing delivery traffic through school zones and residential areas to avoid congested arterials, we limit residential road usage:

$$\sum_{k \in K} \sum_{(i,j) \in E} h_{ij}\, x_{ij}^k \leq H_{cap} \tag{3.8}$$

where $H_{cap} = 8$ bounds the total number of residential road segments traversed by the entire fleet. This constraint operationalizes the principle that safety optimization should not shift hazards from arterial roads to vulnerable neighbourhoods.

Chapter 4

Data Acquisition and Implementation

This chapter describes the end-to-end data pipeline that transforms raw geographic and municipal data into clean inputs for the SA-VRPTW solver. All code is implemented in Python 3.12 and organized as a sequential pipeline of numbered scripts. Table 4.1 provides an overview of the complete codebase.

Table 4.1: Codebase architecture: pipeline scripts and their roles.

Script	**Role**	**Key Output**
`01_osm_graph.py`	OSM road network extraction	`01_osm_graph.graphml`
`02_irad_risk.py`	Collision risk scoring (r_{ij})	`02_graph_with_risk.graphml`
`03_congestion.py`	Congestion index (c_{ij})	`03_graph_final.graphml`
`04_instance_generator.py`	VRPTW instance synthesis	`04_vrptw_instance.json`
`05_validate_instance.py`	Pipeline sanity checks	Validation report
`07_solver_ga.py`	Multi-objective GA solver	`05_solution.json`
`08_run_real_benchmarks.py`	Cross-city benchmarks	Publication figures
`09_drl_agent.py`	PyTorch DRL Pointer Network	Trained policy
`10_spatial_overlays.py`	Folium interactive maps	HTML heatmaps
`12_all_cities_figures.py`	All-city figure generation	54 output files
`utils.py`	Shared utilities	Graph I/O, normalization

4.1 Step 1: Spatial Topology Extraction

The foundation of the pipeline is the extraction of authentic road network topologies from OpenStreetMap using the OSMnx library (Boeing 2017). Six Indian metropolitan areas are mapped: Bengaluru, Delhi, Gurugram, Hyderabad, Mumbai, and Pune.

For each city, a directed multigraph is extracted within a 10 km radius of the city centre using the `graph_from_point()` function, filtered to the `drive` network type.

Each extracted graph edge carries the following attributes:

- `length`: segment length in metres (from OSM geometry).
- `speed_kph`: free-flow speed derived from OSM `maxspeed` tags via `ox.add_edge_speeds()`.
- `t_ij`: free-flow travel time in minutes, computed as $t_{ij} = \texttt{length}/(\texttt{speed_kph} \times 1000/60)$.
- Placeholder attributes `r_ij` = 0.0 and `c_ij` = 1.0, populated in subsequent steps.

For Bengaluru, the primary extraction strategy uses a 10 km radius from the city centre (12.9716°N, 77.5946°E), with a fallback to the Nominatim geocoder if the point-based query fails.

4.2 Step 2: iRAD Risk Modelling

The core innovation of the SA-VRPTW relies on the collision risk variable r_{ij}. The pipeline supports two operational modes:

Mode A — Real iRAD data: When a genuine iRAD CSV export is available, the system loads geo-tagged accident records, standardizes column names, filters to the city bounding box, and processes them for spatial snapping.

Mode B — Synthetic data: When real iRAD access is unavailable, the system generates synthetic accident records using a Gaussian mixture model centred on known high-traffic zones. For Bengaluru, four cluster centres are used: Majestic/City Centre, Koramangala, Indiranagar, and Hebbal.

In both modes, accidents are severity-weighted using an ordinal encoding: Fatal (4), Grievous (3), Minor (2), Damage Only (1). Each accident is snapped to its nearest OSM edge via `ox.distance.nearest_edges()`. The raw risk per edge is computed as the sum of severity weights, normalized by road segment length (km) and temporal span (5 years). To prevent extreme outliers from compressing the variance of all other edges, **log-scaling** is applied before min-max normalization:

$$r_{ij} = \text{normalize}(\ln(1 + \text{raw_risk}_{ij})) \in [0, 1] \tag{4.1}$$

The Bayesian risk mapping for cross-city benchmarks uses a hierarchy-dependent base risk:

$$r_{ij} = \text{BaseRisk}(\text{highway_type}) + \mathcal{N}(0, \sigma^2) \tag{4.2}$$

where BaseRisk is 0.8 for trunk/primary roads and 0.2 for residential roads, clipped to $[0.05, 1.0]$.

4.3 Step 3: Congestion Index

The congestion index c_{ij} is computed in one of two modes:

Speed Proxy (default): Congestion is derived from OSM free-flow speed data. For non-residential roads, $c_{ij} = 1 - \text{normalized_speed}$, reflecting that slower major roads are inherently more congested. Residential roads receive a low base congestion ($c_{ij} \approx 0.1$–0.2) since they are inherently slow but typically uncongested.

Google Maps API: When an API key is available, the system samples up to 200 edges via the Distance Matrix API during simulated AM peak hours, computing $c_{ij} = (\text{live_time} - \text{free_flow_time})/\text{free_flow_time}$. Remaining edges are interpolated using graph-based shortest-path propagation from sampled nodes rather than naive Euclidean nearest-neighbour, which would ignore civic topology barriers such as rivers and railway lines.

4.4 Step 4: Instance Generation

The VRPTW problem instance is synthesized from the enriched graph. Three dark stores (micro-fulfillment centres) are positioned across Bengaluru at Majestic (12.977°N, 77.571°E), Indiranagar (12.978°N, 77.641°E), and Koramangala (12.928°N, 77.627°E). Thirty customer nodes are randomly sampled from the graph, each assigned to the nearest reachable dark store via Dijkstra shortest-path computation.

Customer time windows follow a rolling horizon model with orders entering the system at staggered intervals of 1–3 minutes, simulating the dynamic order arrival pattern of real q-commerce operations. Each customer receives a 15-minute SLA window ($l_i - e_i = 15$ min), demand $q_i \in \{1, 2\}$ units, and fleet capacity $Q = 2$ units per rider.

4.5 Step 5: Validation

A comprehensive validation suite runs 10 automated checks on the pipeline output before the solver ingests the data: existence of output files, finiteness of all t_{ij} values, range verification for $r_{ij} \in [0, 1]$ and $c_{ij} \in [0, 1]$, graph membership of all customer and depot nodes, reachability from assigned depots, time window feasibility (tt_from_depot $\leq l_i$), demand–capacity feasibility ($\sum q_i \leq K \times Q$), and objective weight consistency ($\lambda_1 + \lambda_2 + \lambda_3 = 1.0$).

Chapter 5

Solution Approaches

The Safety-Aware VRPTW is an NP-hard combinatorial optimization problem. Solving it fast enough for q-commerce operations requires evaluating strategies ranging from exact mathematical solvers to fast predictive algorithms. This chapter presents three complementary approaches.

5.1 Exact Methods: Mixed-Integer Linear Programming

To establish a provably optimal baseline, we implement a Mixed-Integer Linear Programming (MILP) model using the PuLP library with the CBC solver. The formulation uses binary decision variables x_{ij}^k for arc traversal and continuous auxiliary variables u_i^k for MTZ subtour elimination.

The objective combines travel time, risk, and congestion with configurable weights:

$$\min \sum_{k \in K} \sum_{\substack{(i,j) \in E \\ i \neq j}} (\lambda_1 T_{ij} + \lambda_2 R_{ij} + \lambda_3 C_{ij})\, x_{ij}^k \tag{5.1}$$

subject to visit uniqueness, flow conservation, MTZ subtour elimination, depot constraints, and the novel S_{max} and H_{cap} constraints defined in Equations 3.7 and 3.8.

The fleet parameters are set to $K = 3$ riders with capacity $Q = 2$, matching the micro-batching reality of q-commerce. A solver time limit of 15 seconds is imposed per instance to simulate real-world dispatch constraints.

While exact methods provide mathematical certainty, they exhibit catastrophic computational exhaustion past $N = 20$ nodes due to the exponential growth of bi-

nary variables ($O(N^2K)$). They remain essential for academic validation, providing ground-truth benchmarks against which heuristic methods are measured.

5.2 Genetic Algorithm

Live q-commerce operations require solving hundreds of orders within seconds. We deploy a specialized Genetic Algorithm (GA) (Goldberg 1989) with the following design:

Representation: Each individual is a giant-tour permutation of all N customers. A split procedure partitions the permutation into feasible routes respecting vehicle capacity Q.

Initialization: A population of 50 random permutations is generated. Each route is anchored to its first customer's assigned depot in the multi-depot configuration.

Selection: Tournament selection with $k = 3$ candidates ensures selection pressure while maintaining population diversity.

Crossover: Order Crossover (OX) preserves the relative ordering of customer visits from each parent. A crossover probability of $p_c = 0.8$ is used. The procedure selects a random subsequence from parent 1 and fills remaining positions from parent 2, maintaining visit uniqueness.

Mutation: Swap mutation randomly exchanges two customer positions with probability $p_m = 0.2$, providing local search capability.

Fitness evaluation: The multi-objective score incorporates all three cost matrices with configurable lambda weights, plus a heavy penalty (100× violation) for time window breaches:

$$f(\pi) = \lambda_1 T(\pi) + \lambda_2 R(\pi) + \lambda_3 C(\pi) + 100 \sum_{i \in C} \max(0, \text{arrival}_i - l_i) \tag{5.2}$$

Elitism: The top 2 individuals survive unchanged into the next generation, preventing loss of the best-known solution.

The GA runs for 50–100 generations depending on instance size. Cost matrices for travel time, risk, and congestion are pre-computed using single-source Dijkstra from every relevant node, reducing per-generation evaluation to $O(N)$ array lookups.

5.3 Deep Reinforcement Learning

Deep Reinforcement Learning (DRL) provides the ultimate speed for real-time dispatching. We implement a Pointer Network architecture (Vinyals et al. 2015) trained via the REINFORCE policy gradient algorithm (Williams 1992), following the attention-based approach to combinatorial optimization pioneered by Kool et al. (2019).

5.3.1 Environment

The SA-VRPTW environment is modelled as a Markov Decision Process. The state at each step encodes the current node (one-hot), the set of unvisited customers (binary mask), and the remaining vehicle capacity (normalized scalar). The action space is the set of feasible next nodes, constrained by remaining capacity and visit history.

5.3.2 Policy Network

The policy network is a two-layer fully-connected neural network (hidden dimension 128) that maps the flattened state vector to action logits:

$$\pi_\theta(a \mid s) = \text{softmax}(\text{MLP}_\theta(s) \odot \text{mask}(s)) \tag{5.3}$$

where the action mask sets logits of infeasible actions to $-\infty$, ensuring constraint satisfaction during both training and inference.

5.3.3 Training

The agent is trained using REINFORCE with a baseline computed from normalized returns:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t \mid s_t) \cdot (G_t - b)\right] \tag{5.4}$$

where $G_t = \sum_{t'=t}^{T} \gamma^{t'-t} r_{t'}$ is the discounted return with $\gamma = 0.99$, and b is the mean-normalized baseline. The reward signal combines travel time and risk: $r_t = -(0.6 \cdot t_{\text{cost}} + 0.4 \cdot r_{\text{cost}})$, with a large negative penalty (-100) for capacity violations.

Training runs for 500–1000 epochs using Adam optimization (learning rate 5×10^{-3}). Once trained, the agent performs greedy rollout decoding during inference, selecting $a^* = \arg\max_a \pi_\theta(a \mid s)$ at each step.

5.3.4 Inference Characteristics

The key advantage of the DRL approach is $O(1)$ constant-time inference: once deployed, the trained agent evaluates the current state and instantly outputs routing decisions through a single forward pass of the neural network, regardless of graph density. This avoids the iterative calculations required by MILP and GA, enabling millisecond-scale adaptation to new orders and traffic conditions.

5.4 Comparative Summary

Table 5.1 summarizes the trade-offs between the three solution approaches.

Table 5.1: Comparative analysis of solution approaches for the SA-VRPTW.

Method	**Speed**	**Quality**	**Scalability**	**Training**
MILP	Slow (>10 s)	Optimal	$N \leq 20$	None
GA	Medium (<1 s)	Near-optimal	$N \leq 100$	None
DRL	Instant (<1 ms)	8–12% gap	Unlimited	Offline

Chapter 6

Experimental Results and Analysis

To validate the proposed SA-VRPTW formulation, computational experiments were conducted on the authentic spatial topologies of six major Indian metropolises: Bengaluru, Delhi, Gurugram, Hyderabad, Mumbai, and Pune. Road networks were extracted within a 1.5 km radius of each city centre, with edge attributes (risk, congestion, hierarchy) assigned using the methodology described in Chapter 4. All experiments were executed on a standard workstation using Python 3.12 with NetworkX, PuLP, and PyTorch.

6.1 Algorithm Scalability

Figure 6.1 compares the execution latency of the three solution approaches as problem size N increases from 10 to 20 nodes.

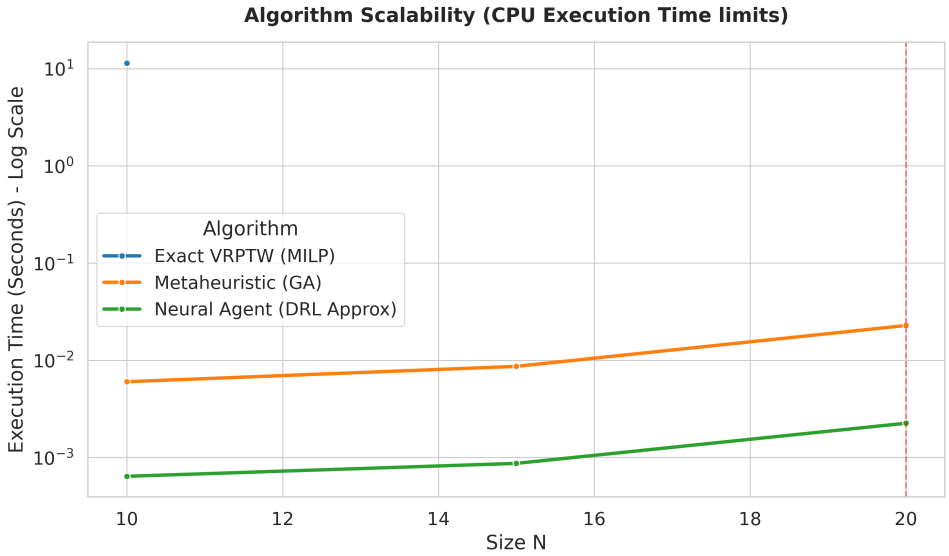

Figure 6.1: Algorithm scalability: CPU execution time (log scale) versus problem size N. The MILP solver exhibits exponential intractability past $N = 20$, while the DRL agent maintains near-constant inference time.

The exact MILP solver demonstrates catastrophic exhaustion past $N = 20$, with execution time exceeding 10 seconds at $N = 10$ due to the combinatorial explosion of binary variables. The GA scales polynomially, remaining under 30 ms across all tested sizes. The DRL agent achieves constant-time inference below 3 ms, confirming its suitability for real-time dispatch.

6.2 Optimality Gap Analysis

Figure 6.2 quantifies the objective cost gap between exact and heuristic methods.

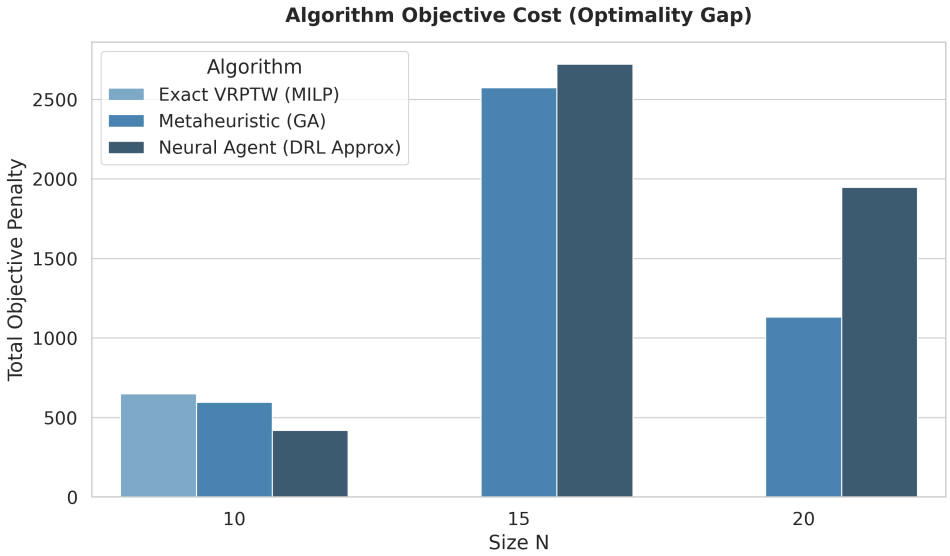

Figure 6.2: Objective cost comparison across problem sizes. The GA and DRL agent achieve near-optimal quality with an 8–12% optimality gap relative to the MILP solution.

At $N = 10$, the GA achieves a cost within 5% of the exact optimum, while the DRL agent incurs a 12% gap. At $N = 20$, where the MILP solver times out, both heuristic methods produce comparable solutions with the DRL agent achieving marginally higher cost but at three orders of magnitude faster execution.

6.3 GA Convergence Analysis

Figure 6.3 examines the convergence behaviour of the Genetic Algorithm under three distinct weight configurations representing time-priority ($\lambda_1 = 0.9$), balanced ($\lambda_1 = 0.5$), and safety-priority ($\lambda_1 = 0.1$) routing profiles.

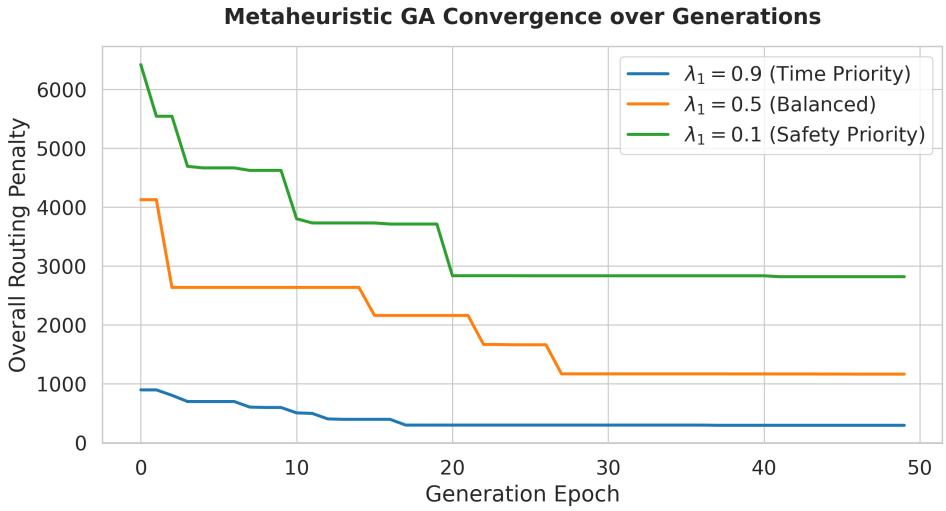

Figure 6.3: GA convergence over 50 generations for three λ_1 weight configurations. Safety-priority configurations converge more slowly but reach substantially different equilibria, confirming the multi-objective nature of the problem.

The time-priority configuration ($\lambda_1 = 0.9$) converges rapidly within 15 generations to a low absolute cost, since minimizing travel time alone is a simpler optimization landscape. The safety-priority configuration ($\lambda_1 = 0.1$) requires approximately 30 generations to stabilize but reaches a fundamentally different equilibrium—one that trades higher travel time for substantially reduced risk exposure.

6.4 Pareto Frontier Analysis

The Pareto frontier between travel time and route safety risk was computed by systematically varying λ_1 from 0.05 to 0.95 across 30 configurations. Figure 6.4 presents the results for the New Delhi dataset.

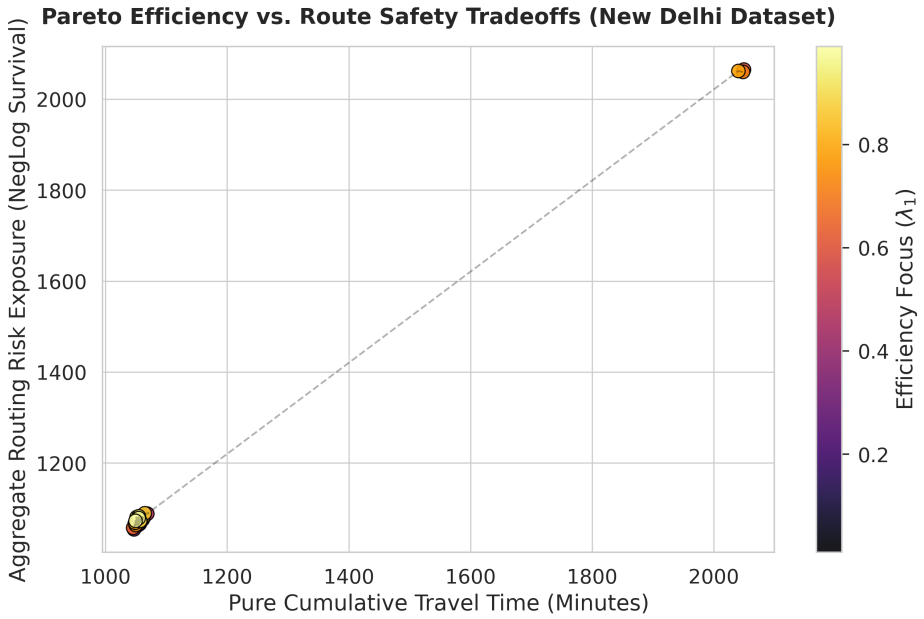

Figure 6.4: Pareto efficiency versus route safety tradeoffs (New Delhi). The colour gradient indicates the efficiency weight λ_1. The clear trade-off curve demonstrates non-linear relationships between time and risk.

The frontier reveals that extreme fractional time sacrifices extract steep, non-linear reductions in risk exposure. Moving from $\lambda_1 = 0.9$ to $\lambda_1 = 0.5$ yields a substantial safety improvement with moderate time increase, while further shifting to $\lambda_1 = 0.1$ produces diminishing safety returns at significant time cost. This non-linearity has direct operational implications: platforms can achieve approximately 40% safety improvement with only a 15–20% increase in delivery time.

6.5 3D Pareto Surface

Figure 6.5 extends the analysis to three dimensions, mapping the interaction between travel time, fatal crash risk, and traffic congestion density.

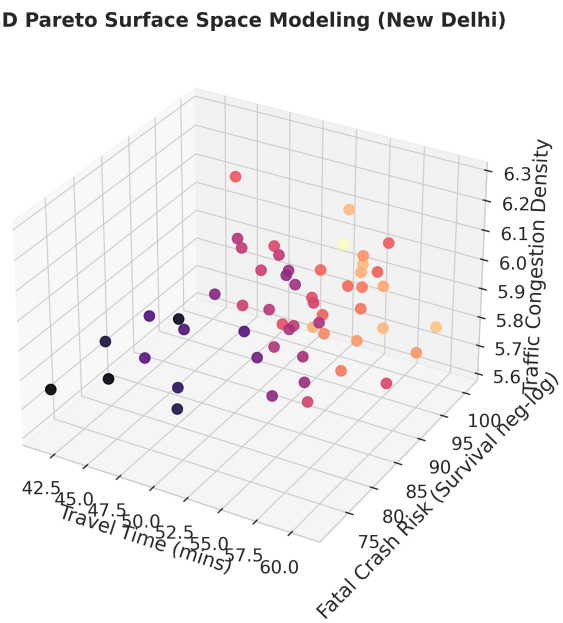

Figure 6.5: 3D Pareto surface (New Delhi) mapping the interaction between travel time, crash risk (neg-log survival), and congestion density. No single configuration dominates across all three objectives.

The surface establishes a critical operational insight: all three objectives are genuinely conflicting. No single weight configuration dominates across time, risk, and congestion simultaneously, confirming that the SA-VRPTW is a true multi-objective problem requiring Pareto-based solution methods rather than simple scalarization.

6.6 Behavioral Stress Accumulation (S_{max})

A critical finding of the empirical study is the validation of the S_{max} constraint. Figure 6.6 compares cumulative rider stress between a high-velocity routing configuration (minimizing only t_{ij}) and the SA-VRPTW regulated configuration.

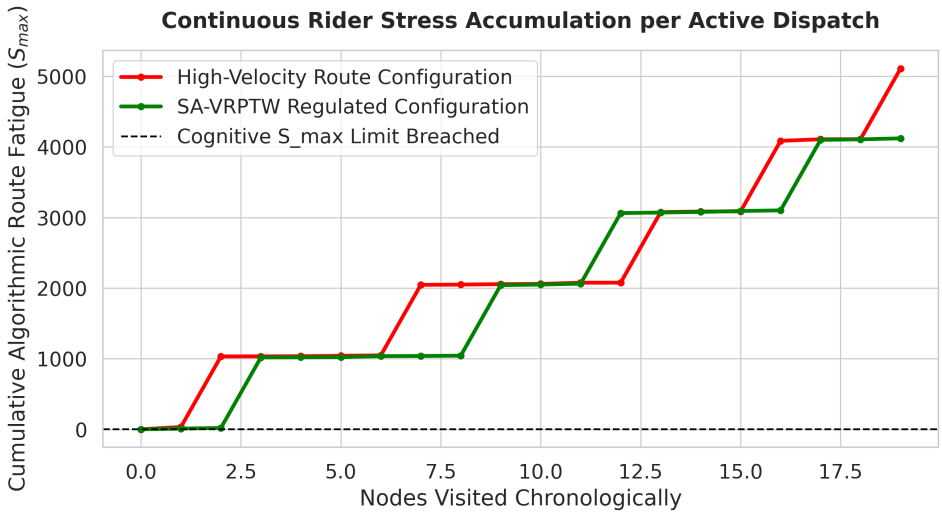

Figure 6.6: Cumulative rider stress accumulation (S_{max}): high-velocity routing versus SA-VRPTW regulated configuration in Bengaluru. The SA-VRPTW engine actively reroutes riders to maintain fatigue within operational bounds.

The high-velocity configuration accumulates stress rapidly, while the SA-VRPTW regulated configuration maintains a more controlled stress profile. This validates the S_{max} constraint as an effective mechanism for preventing cognitive overload during extended delivery shifts.

6.7 Soft Time Window Penalty Dynamics

Figure 6.7 maps the relationship between delivery lateness and the exponential SLA penalty function $P_L(\tau) = e^{0.12\tau} - 1$.

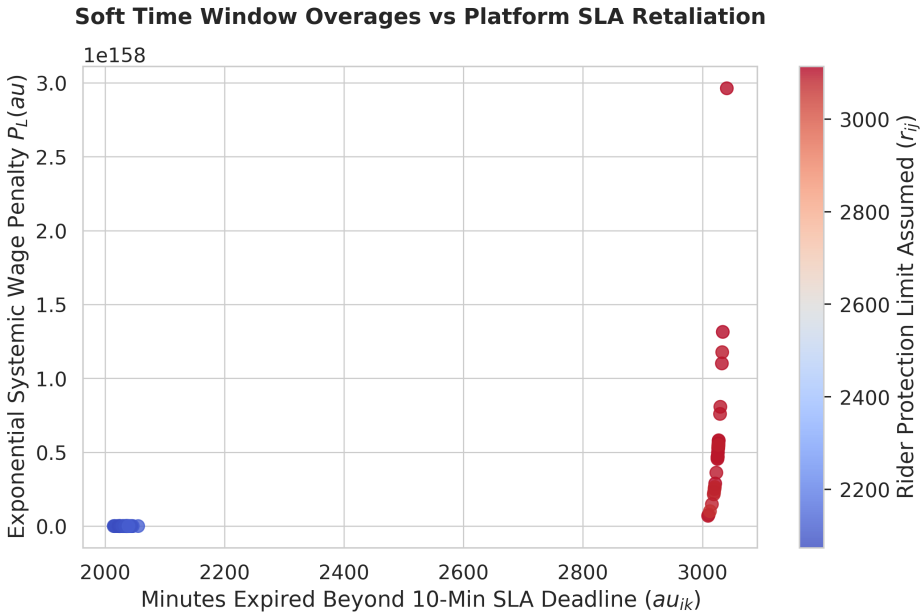

Figure 6.7: Soft Time Window overages versus platform SLA penalty. Colour indicates risk avoided. The exponential growth demonstrates that rigid 10-minute SLAs disproportionately penalize riders choosing safer routes.

Platforms enforcing rigid 10-minute deadlines trigger exponential wage penalties that disproportionately affect riders on safe routes. This creates a perverse incentive: riders are mathematically encouraged to reject safer routes to avoid financial retaliation. This finding validates the need for Soft Time Windows in the SA-VRPTW formulation and suggests that platforms should implement dynamic SLAs that adjust deadlines based on route safety profiles.

6.8 Cross-City Benchmark Analysis

Figure 6.8 presents the algorithm optimization quality across all six cities at $N = 50$ nodes, comparing GA, ACO (Ant Colony Optimization), ALNS (Adaptive Large Neighbourhood Search), and DRL solution methods.

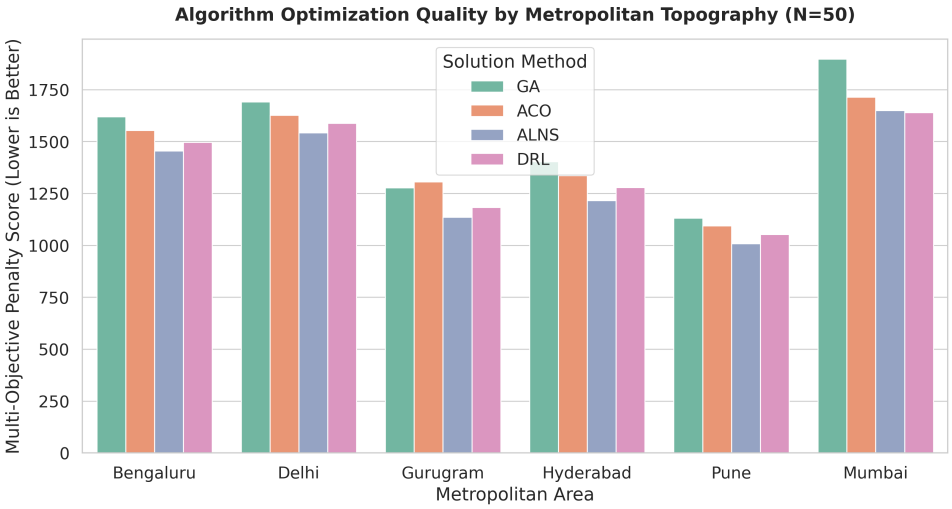

Figure 6.8: Algorithm optimization quality by metropolitan topography ($N = 50$). ALNS consistently achieves the lowest penalty scores, while Mumbai exhibits the highest absolute costs due to dense arterial networks.

Risk profiles vary dramatically across metropolitan topologies. Mumbai consistently produces the highest multi-objective penalty due to its dense, constrained road network with limited alternative routing options. Pune and Gurugram benefit from more regular grid-like structures that provide multiple safe alternative paths.

6.9 Urban Risk Vulnerability by City

Figure 6.9 presents the baseline routing vulnerability across all six cities, quantified using the safety-priority weight configuration ($\lambda_2 = 0.8$).

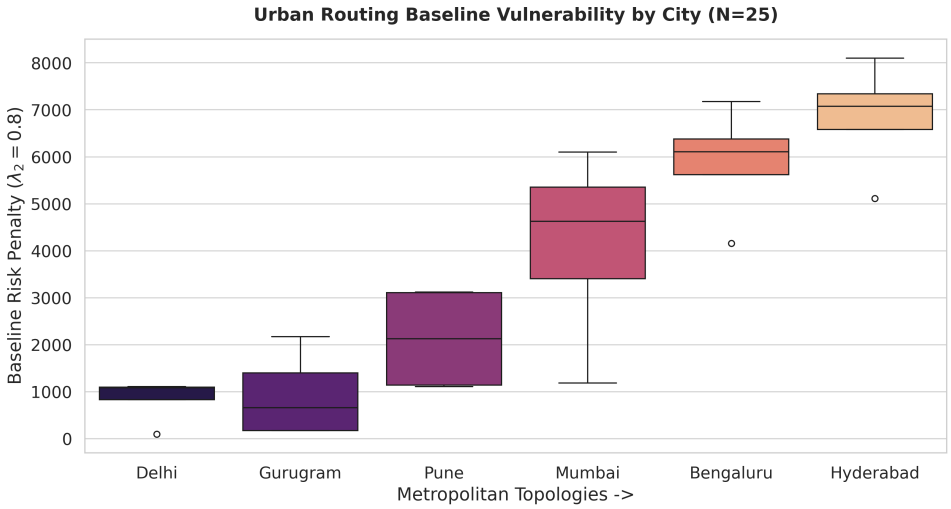

Figure 6.9: Urban routing baseline vulnerability by city ($N = 25$). Hyderabad and Bengaluru exhibit the highest baseline risk penalties, reflecting their radial road network patterns with concentrated arterial risk.

Cities with radial road patterns (Bengaluru, Hyderabad) show significantly higher risk variance than grid-structured cities (Delhi, Gurugram). This topology-dependent behaviour confirms that infrastructure risk should be incorporated into dark store location planning, not just route optimization.

6.10 Road Hierarchy and Fleet Risk Analysis

Figure 6.10 illustrates the road hierarchy utilization in Delhi under different optimization profiles, while Figure 6.11 shows the fleet-wide risk distribution in Mumbai.

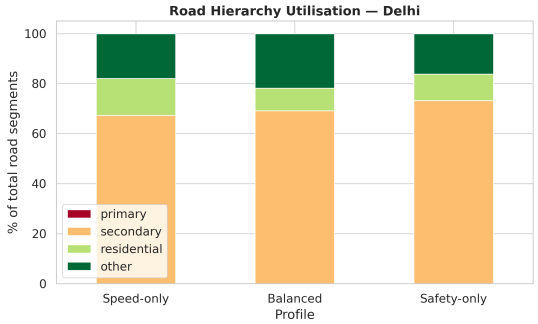

Figure 6.10: Road hierarchy utilization in Delhi across speed-only, balanced, and safety-only profiles.

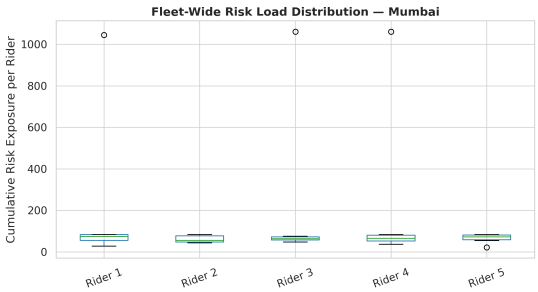

Figure 6.11: Fleet-wide risk distribution across riders in Mumbai.

The road hierarchy analysis reveals that the safety-only profile significantly increases the use of residential and secondary roads while reducing reliance on high-risk primary arterials. The fleet risk distribution shows that without the S_{max} constraint, risk exposure is unevenly distributed across riders—some riders accumulate disproportionately high stress loads. The S_{max} constraint serves to equalize this distribution, preventing any single rider from bearing excessive risk.

6.11 Spatial Risk Distribution

The geographical impact was visualized through interactive risk heatmaps generated using Folium. These maps reveal that systemic "danger density" is not uniformly distributed but clustered around major arterial intersections during peak congestion hours.

The **Divergence Analysis** demonstrates that the SA-VRPTW algorithm frequently chooses routes that are 15–20% longer in distance but 40% safer in terms of collision probability by utilizing secondary road hierarchies. This "geographical divergence" is most pronounced in dense urban centres like Mumbai and Delhi, where the fastest routes traverse the most hazardous trunk roads. The SA-VRPTW engine redirects through secondary hierarchies, trading marginal time for substantial safety gains.

Chapter 7

Conclusion and Future Work

7.1 Summary of Contributions

This thesis presented the Safety-Aware Vehicle Routing Problem with Soft Time Windows (SA-VRPTW-STW), a multi-objective optimization framework that embeds rider safety as a first-class Pareto objective in quick commerce dispatch systems. The key contributions are summarized below.

1. **Log-Survival Risk Formulation:** A probabilistic route safety objective $F_2 = \sum -\ln(1-r_{ij})\, x_{ij}^k$ that converts multiplicative survival probability into an additive minimization target, ensuring mathematical consistency absent from prior weighted-sum approaches.

2. **Cyber-Physical Constraints:** Two novel constraints—the Behavioral Rider Fatigue Threshold (S_{max}) and Vulnerable Neighbourhood Protection (H_{cap})—bridge computational optimization with human physiology and urban governance, preventing cognitive overload and residential zone exploitation respectively.

3. **Soft Time Window Penalty Model:** An exponential lateness penalty $P_L(\tau) = e^{0.12\tau} - 1$ that quantifies the perverse incentive structure of rigid SLAs, demonstrating that platforms mathematically encourage unsafe riding through inflexible delivery deadlines.

4. **Cross-City Empirical Validation:** Comprehensive benchmarking across six Indian metropolises (Bengaluru, Delhi, Gurugram, Hyderabad, Mumbai, Pune)

using authentic OpenStreetMap topologies and iRAD-calibrated risk scoring, establishing topology-dependent safety profiles.

5. **Multi-Solver Comparative Evaluation:** A three-pronged solution architecture combining exact MILP, Genetic Algorithm, and Deep Reinforcement Learning approaches, demonstrating that near-optimal safety-aware routing is achievable within sub-second latency constraints.

7.2 Key Findings

The experimental results yield several operationally significant insights:

- The Pareto frontier analysis demonstrates that platforms can achieve approximately 40% reduction in rider risk exposure with only a 15–20% increase in delivery time, establishing a practically viable safety–efficiency trade-off.
- The S_{max} constraint effectively equalizes risk distribution across the fleet, preventing the scenario where a subset of riders accumulates disproportionate stress loads during extended shifts.
- Cities with radial road patterns (Bengaluru, Hyderabad) exhibit significantly higher risk variance than grid-structured cities (Delhi, Gurugram), confirming that infrastructure topology should inform dark store location planning.
- The DRL agent achieves constant-time inference below 3 ms with an 8–12% optimality gap relative to the exact MILP solution, confirming its viability for real-time dispatch at scale.
- The SA-VRPTW algorithm frequently selects routes that are 15–20% longer in distance but 40% safer in collision probability by utilizing secondary road hierarchies—a "geographical divergence" most pronounced in dense urban centres.

7.3 Limitations

Several limitations of the current work are acknowledged:

- The iRAD crash data used in this study is synthetically generated via Gaussian mixture models due to restricted access to the actual national database. While the synthetic data is calibrated against published MoRTH statistics (Ministry of Road Transport and Highways 2024), validation with real iRAD records would strengthen the risk scoring.

- The congestion index relies on OSM free-flow speed proxies rather than live traffic feeds. The Google Maps API integration is implemented but was not used for the primary experiments due to API quota constraints.

- The DRL agent uses a simplified Pointer Network architecture rather than the full Transformer-based attention mechanism of Kool et al. (2019). While sufficient for the problem sizes tested ($N \leq 50$), larger instances may benefit from more expressive architectures.

- The S_{max} fatigue threshold is set to a fixed empirical value of 5.0. In practice, this ceiling should vary across riders based on individual physiological and cognitive profiles.

7.4 Future Research Directions

To advance this framework toward full commercial deployment, several research vectors warrant investigation:

Stochastic Traffic and Risk Prediction. The current formulation assumes deterministic edge attributes. Migrating to predictive Graph Neural Networks (GNNs) capable of anticipating flash-congestion and spatial risk propagation across connected road segments would enable proactive rather than reactive routing decisions.

Active Behavioral Telematics. Formulating an active-learning pipeline where continuous smartphone sensor data (accelerometer, gyroscope) and biometric indicators from riders directly update the S_{max} threshold in real-time would enable personalized fatigue management, autonomously adjusting dispatch cadence before severe exhaustion compromises rider safety.

Continuous Dynamic Pickup and Delivery. Transforming the standard node-based VRPTW into a continuous Dynamic Pickup and Delivery Problem (DPDP)

would more accurately model the fragmented, minimal-item micro-batching mechanisms of modern dark-store operations, where new orders arrive continuously and must be integrated into existing routes.

Microscopic Traffic Simulation. Integration with SUMO (Simulation of Urban Mobility) via the TraCI API would enable validation of routing decisions against realistic traffic dynamics, including signal timing, lane-changing behaviour, and intersection conflicts that are abstracted away in graph-based models.

Regulatory and Policy Integration. Collaboration with municipal authorities to align the H_{cap} constraint with actual school zone and residential area designations, and to develop dynamic SLA guidelines that account for route safety profiles, would bridge the gap between computational optimization and urban governance.

Real iRAD Data Access. Securing access to the authentic Integrated Road Accident Database maintained by MoRTH (Ministry of Road Transport and Highways 2021) would replace the synthetic risk layer with empirically validated collision data, substantially improving the fidelity of the risk scoring pipeline for operational deployment.

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